

Ramansh Sharma

CONTACT INFORMATION	72 Central Campus Dr WEB 2780 Salt Lake City, UT 84112 USA	<i>Voice:</i> (801) 414-6734 <i>E-mail:</i> ramansh@cs.utah.edu <i>url:</i> https://rsmath.github.io
RESEARCH INTERESTS	Neural operator architectures (DeepONet, FNO, Kernels) for interpretable, scalable, and adaptive operator learning and physics-informed learning for scientific machine learning applications.	
EDUCATION	University of Utah , Salt Lake City, Utah USA, 2023-2028 (expected) Ph.D. in Computing Advisor: Varun Shankar GPA: 3.8/4 SRM University , Chennai, Tamil Nadu India, 2019-2023 Bachelors of Technology, Computer Science and Engineering GPA: 9.7/10 (3.9/4)	
HONORS AND AWARDS	• Graduated first class with distinction from SRM University, 2023 • NeurIPS Scholar Award, 2022 • NeurIPS Travel Award, 2022	
ACADEMIC EXPERIENCE	University of Utah , Salt Lake City, UT USA <i>Graduate Student</i> August, 2023 - Present Ongoing Ph.D. research on operator learning methods for scientific machine learning applications. <i>Teaching Assistant</i> August, 2024 - December 2024 CS 4230/6230 (Parallel and High-Performance Computing)	
PUBLICATIONS	Google Scholar - https://scholar.google.com/citations?hl=en&user=lUmqHckAAAAJ 1. Ramansh Sharma, Matthew Lowery, Houman Owhadi, and Varun Shankar. Fluids You Can Trust: Property-Preserving Operator Learning for Incompressible Flows (Preprint). 2. Ramansh Sharma and Varun Shankar. Ensemble and Mixture-of-Experts DeepONets For Operator Learning (TMLR, 2025). 3. Ramansh Sharma and Varun Shankar. Accelerated Training of Physics Informed Neural Networks (PINNs) using Meshless Discretizations (NeurIPS, 2022). 4. Jordi Planas, Daniel F. Quevedo, Galina Naydenova, Ramansh Sharma, Cristina Taylor, Kathleen Buckingham, and Rong Fang. Beyond modeling: NLP Pipeline for efficient environmental policy analysis (KDD, 2021).	
SERVICE	Vice-President of Graduate Student Advisory Committee.	August, 2024 - August 2025
REVIEWING EXPERIENCE	1. Neural Information Processing Systems (NeurIPS).	

PROFESSIONAL
EXPERIENCE

Kahlert School of Computing, University of Utah

Research Assistant

August, 2023 - Present

Working on interpretable, scalable, and adaptive operator learning methods for a variety of scientific machine learning applications.

Los Alamos National Laboratory

Graduate Student

May, 2025 - August 2025

Worked on developing a scale-bridging framework that connects fine-scale pore-level flow physics (Stokes equations) to coarse-scale models (Darcy and Brinkman equations) using numerical solvers and machine learning. By combining inverse modeling, differentiable solvers, and data-driven parameter learning, it aims to efficiently predict macroscopic flow behavior from microscopic structures in porous media.

Approximate Bayesian Inference team, RIKEN

Remote Collaborator

October, 2021 - June, 2023

Research focusing on **curriculum learning** and its advantages over independent and identically distributed (**i.i.d.**) training. Implemented and executed comprehensive experiments with memorability metrics such as **residual** and **leverage scores**. Presented technical reports with the methodology and results to the team.

World Resources Institute

Machine Learning Engineer

February, 2021 - September, 2021

Identifying economic and financial incentives for forest and landscape restoration using Natural Language Processing. Implemented custom **early stopping** features for **sentence transformers**. Led the experiments and discussion around the reproducibility issue in policy instrument **binary** and **multiclass classification** with **Sentence-BERT**. Presented our team's paper, *Beyond modeling: NLP Pipeline for efficient environmental Policy Analysis*, at ACM's annual KDD conference at the Fragile Earth workshop.

Omdena

Platform Engineer

January, 2021 - September, 2021

Developed Airtable scripts in Javascript to index, modify, and maintain the community database. Developed an end-to-end system for community engagement and participation by integrating GitHub, Slack, and Airtable APIs together. Designed the first of its kind AI-4-good Python library, Omde-naLore, with more than 35000 lines of code.

Data Scientist

October, 2020 - December, 2020

Worked for a silicon valley startup, connecting volunteer opportunities with prospective volunteers in a timebanking system. Designed multiple recommender models for volunteers and project administrators based on skills and interests.

Machine Learning Engineer

August, 2020 - September, 2020

Analyzed the environmental, geographic, geospatial, and socio-economic factors that contributed to illegal dumpsites around the world. Led the ML team through modeling, iterating, and refining classifiers to predict illegal dumpsites based on environmental, geographic, geospatial, and socio-economic factors factors such as population, population density, distance to roads, venue categories, and distance to venue categories.

GRADUATE
COURSEWORK

1. Graduate Algorithms
2. High Dimensional Data Analysis
3. Scientific Computing I
4. Scientific Computing II

5. Probabilistic Machine Learning
6. High Performance & Parallel Computing

COMPUTER SKILLS

- Languages: Python (**8** years experience), Matlab (Octave), C++, Java, Javascript.
- Parallel Programming: CUDA, MPI, OpenMP.
- Machine Learning: PyTorch, Jax, Haiku, NNX, Optax, CuPy, TensorFlow.
- Technologies: Weights&Biases, Flask, Heroku, Docker, FastAPI.
- Scientific Visualization: Paraview.